RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

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RAJALAKSHMI ENGINEERING COLLEGE

Pranavkrishna V S 250701540-2025-CSE P2

Started on	Tuesday, 25 August 2026, 10:32 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 10:34 AM
Time taken	2 mins 47 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    int i = 1;
    int s = 1;
    int count = 0;

    scanf("%d", &n);

    while (s <= n)
    {
        count++;
        i++;
        count++;
        s += i;
        count++;
    }

    count += 3;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, count = 0;

    scanf("%d", &n);

    if (n == 1)
    {
        count = 1;
    }
    else
    {
        count = 5 * n + 2;
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int num, i;
    int count = 0;

    scanf("%d", &num);

    for (i = 1; i <= num; ++i)
    {
        count++;  // i <= num

        if (num % i == 0)
        {
            count++;  // printf executes
        }
    }

    count += num;  // ++i executed num times
    count++;       // final loop condition

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔

	Input	Expected	Got	
✔	4	12	12	✔

Passed all tests! ✔

Correct

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Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int c = 0;
    int i, j, k;

    scanf("%d", &n);

    c++;  // initialization of i

    for (i = n / 2; ; i++)
    {
        c++;  // i < n condition

        if (i >= n)
            break;

        for (j = 1; ; j = 2 * j)
        {
            c++;  // j < n condition

            if (j >= n)
                break;

            for (k = 1; ; k = k * 2)
            {
                c++;  // k < n condition

                if (k >= n)
                    break;

                c++;  // c++
            }
        }
    }
```

```
39      printf("%d", c);
40
41      return 0;
42  }
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, temp, digits = 0;
    int count;

    scanf("%d", &n);

    temp = n;

    while (temp != 0)
    {
        digits++;
        temp /= 10;
    }

    if (digits == 1)
        count = 9;
    else
        count = 4 * digits + 3;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔

	Input	Expected	Got	
✔	1234	19	19	✔

Passed all tests! ✔

Correct

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Question 1 | Correct Mark 1.00 out of 1.00

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int V, count = 0;
    int denominations[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};
    int n = 9;

    scanf("%d", &V);

    for (int i = 0; i < n; i++) {
        count += V / denominations[i];
        V = V % denominations[i];
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	49	5	5	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input:

3

1 2 3

2

1 1

Output:

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

Constraints:

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    long long x = *(long long *)a;
    long long y = *(long long *)b;

    if (x < y) return -1;
    if (x > y) return 1;
    return 0;
}

int main() {
    int n, m;

    // Number of children
    scanf("%d", &n);

    long long g[n];

    // Greed factors
    for (int i = 0; i < n; i++) {
        scanf("%lld", &g[i]);
    }

    // Number of cookies
    scanf("%d", &m);

    long long s[m];

```

```
    // Cookie sizes
    for (int i = 0; i < m; i++) {
        scanf("%lld", &s[i]);
    }

    // Sort greed factors and cookie sizes
    qsort(g, n, sizeof(long long), compare);
    qsort(s, m, sizeof(long long), compare);

    int child = 0;
    int cookie = 0;
    int count = 0;

    // Greedy assignment
    while (child < n && cookie < m) {
        if (s[cookie] >= g[child]) {
            count++;
            child++;
            cookie++;
        } else {
            cookie++;
        }
```

	Input	Expected	Got	
✔	2 1 2 3 1 2 3	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3

burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance

he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

Input Format

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

```
3
5 10 7
```

Sample Output

```
76
```

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<math.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] < arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    int s=0;
    for(int i=0;i<n;i++){
        s+=(pow(n,i))*arr[i];
```

```
    }
    printf("%d",s);
}
```

	Test	Input	Expected	Got	
✔	Test Case 1	3 1 3 2	18	18	✔
✔	Test Case 2	4 7 4 9 6	389	389	✔
✔	Test Case 3	3 5 10 7	76	76	✔

Passed all tests! ✔

Correct

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Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    int x = *(int *)a;
    int y = *(int *)b;

    return (x > y) - (x < y);
}

int main() {
    int n;
    scanf("%d", &n);

    long long arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%lld", &arr[i]);
    }

    // Sort in ascending order
    qsort(arr, n, sizeof(long long), compare);

    long long sum = 0;

    // Calculate maximum sum
    for (int i = 0; i < n; i++) {
        sum += arr[i] * i;
    }

    printf("%lld\n", sum);

    return 0;
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int ascending(const void *a, const void *b) {
    return (*(int *)a > *(int *)b) - (*(int *)a < *(int *)b);
}

int descending(const void *a, const void *b) {
    return (*(int *)b > *(int *)a) - (*(int *)b < *(int *)a);
}

int main() {
    int n;
    scanf("%d", &n);

    long long A[n], B[n];

    for (int i = 0; i < n; i++)
        scanf("%lld", &A[i]);

    for (int i = 0; i < n; i++)
        scanf("%lld", &B[i]);

    qsort(A, n, sizeof(long long), ascending);
    qsort(B, n, sizeof(long long), descending);

    long long sum = 0;

    for (int i = 0; i < n; i++)
        sum += A[i] * B[i];

    printf("%lld\n", sum);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int floorValue(int a[], int l, int r, int x)
{
    if (l > r)
        return -1;

    int mid = (l + r) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] > x)
        return floorValue(a, l, mid - 1, x);

    int f = floorValue(a, mid + 1, r, x);
    return (f == -1) ? a[mid] : f;
}

int main()
{
    int n, x;
    scanf("%d", &n);

    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d", floorValue(a, 0, n - 1, x));

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

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Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int countZero(int a[], int l, int r)
{
    if (l > r)
        return 0;

    int mid = (l + r) / 2;

    if (a[mid] == 0)
        return (r - mid + 1) + countZero(a, l, mid - 1);

    return countZero(a, mid + 1, r);
}

int main()
{
    int m;
    scanf("%d", &m);

    int a[m];

    for (int i = 0; i < m; i++)
        scanf("%d", &a[i]);

    printf("%d", countZero(a, 0, m - 1));

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔

	Input	Expected	Got	
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

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Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element.*

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int majority(int a[], int l, int r)
{
    if (l == r)
        return a[l];

    int mid = (l + r) / 2;
    int x = majority(a, l, mid);
    int y = majority(a, mid + 1, r);

    if (x == y)
        return x;

    int cx = 0, cy = 0;
    for (int i = l; i <= r; i++) {
        if (a[i] == x) cx++;
        if (a[i] == y) cy++;
    }

    return cx > cy ? x : y;
}

int main()
{
    int n;
```

```
    scanf("%d", &n);

    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("%d", majority(a, 0, n - 1));

    return 0;
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Pranavkrishna V S 250701540-2025-CSE P2

Started on	Tuesday, 25 August 2026, 10:32 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 10:34 AM
Time taken	2 mins 47 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findPair(int a[], int l, int r, int x, int *p, int *q)
{
    if (l >= r)
        return 0;

    int sum = a[l] + a[r];

    if (sum == x) {
        *p = a[l];
        *q = a[r];
        return 1;
    }

    if (sum < x)
        return findPair(a, l + 1, r, x, p, q);

    return findPair(a, l, r - 1, x, p, q);
}

int main()
{
    int n, x, p, q;
    scanf("%d", &n);

    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    if (findPair(a, 0, n - 1, x, &p, &q)) {
        printf("%d\n%d", p, q);
    } else {
        printf("No");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void quicksort(int a[], int low, int high)
{
    if (low >= high)
        return;

    int i = low, j = high;
    int pivot = a[(low + high) / 2];
    int temp;

    while (i <= j)
    {
        while (a[i] < pivot)
            i++;

        while (a[j] > pivot)
            j--;

        if (i <= j)
        {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
            i++;
            j--;
        }
    }

    quicksort(a, low, j);
    quicksort(a, i, high);
}

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    quicksort(a, 0, n - 1);

    for (int i = 0; i < n; i++)
```

```
47  {
48      printf("%d", a[i]);
49      if (i < n - 1)
50          printf(" ");
51  }
52
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 10.00 out of 10.00

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

Input Format

First Line contains the integer n

The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Helper macro to find the maximum of two numbers
#define MAX(a, b) ((a) > (b) ? (a) : (b))

long long getMaxMonetaryPath(int n, long long grid[n][n]) {
    long long dp[n][n];

    // Base case: Starting position
    dp[0][0] = grid[0][0];

    // Initialize the first row (can only be reached by moving Right)
    for (int j = 1; j < n; j++) {
        dp[0][j] = dp[0][j - 1] + grid[0][j];
    }

    // Initialize the first column (can only be reached by moving Down)
    for (int i = 1; i < n; i++) {
        dp[i][0] = dp[i - 1][0] + grid[i][0];
    }

    // Fill the rest of the DP table
    for (int i = 1; i < n; i++) {
        for (int j = 1; j < n; j++) {
            // Ram takes the maximum value coming from either Up or Left
            dp[i][j] = MAX(dp[i - 1][j], dp[i][j - 1]) + grid[i][j];
        }
    }

```

```
    // The bottom-right cell contains the maximum monetary value
    return dp[n - 1][n - 1];
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    long long grid[n][n];
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%lld", &grid[i][j]);
        }
    }

    printf("%lld\n", getMaxMonetaryPath(n, grid));

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 4 2 3 4 8 7 1	19	19	✔
✔	3 1 3 1 1 5 1 4 2 1	12	12	✔
✔	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*

Output:*6*

Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

long long countWays(int n) {
    if (n < 0) return 0;

    // Using long long array to prevent 32-bit integer overflow
    long long dp[n + 1];

    // Base case
    dp[0] = 1;

    for (int i = 1; i <= n; i++) {
        dp[i] = 0;

        if (i >= 1) {
            dp[i] += dp[i - 1];
        }

        if (i >= 3) {
            dp[i] += dp[i - 3];
        }
    }

    return dp[n];
}
```

```
27 int main() {
28     int n;
29     if (scanf("%d", &n) != 1) return 0;
30
31     // Use %lld format specifier to print a 64-bit long long integer
32     printf("%lld\n", countWays(n));
33
34     return 0;
35 }
36
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

// Helper macro to find the maximum of two numbers
#define MAX(a, b) ((a) > (b) ? (a) : (b))

int getLCSLength(const char *s1, const char *s2) {
    int m = strlen(s1);
    int n = strlen(s2);

    // Create a 2D DP array with extra row and column for empty prefix cases
    int dp[m + 1][n + 1];

    // Initialize base cases where either string prefix is empty
    for (int i = 0; i <= m; i++) {
        dp[i][0] = 0;
    }
    for (int j = 0; j <= n; j++) {
        dp[0][j] = 0;
    }

    // Fill the DP table
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            // If characters match, increment the result from the previous prefixes
            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                // If they don't match, take the maximum possible value by skipping one c
                dp[i][j] = MAX(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }

    // The bottom-right cell contains the length of the longest common subsequence
    return dp[m][n];
```

```
    return dp[m][n];
}

int main() {
    char s1[1001];
    char s2[1001];

    // Read the two strings safely
    if (scanf("%1000s", s1) != 1) return 0;
    if (scanf("%1000s", s2) != 1) return 0;

    printf("%d\n", getLCSLength(s1, s2));

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Function to find the index of the first element greater than target
int upper_bound(int arr[], int size, int target) {
    int low = 0, high = size;
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] <= target) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int lengthOfLNDS(int seq[], int n) {
    if (n == 0) return 0;

    // tails[i] stores the smallest tail of all non-decreasing subsequences of length
    int tails[n];
    int len = 0;

    for (int i = 0; i < n; i++) {
        // Find where seq[i] fits in the tails array
        int idx = upper_bound(tails, len, seq[i]);

        // Update the tail value at that index
        tails[idx] = seq[i];

        // If it extends beyond the current maximum length, increment len
        if (idx == len) {
            len++;
        }
    }

    return len;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int seq[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &seq[i]);
    }

    printf("%d\n", lengthOfLNDS(seq, n));

    return 0;
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findDuplicate(const int arr[], int n) {
    // Phase 1: Finding the intersection point in the cycle
    int slow = arr[0];
    int fast = arr[0];

    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);

    // Phase 2: Finding the entrance to the cycle (the duplicate number)
    int slow2 = arr[0];
    while (slow != slow2) {
        slow = arr[slow];
        slow2 = arr[slow2];
    }

    return slow;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int duplicate = findDuplicate(arr, n);
    printf("%d\n", duplicate);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findDuplicate(const int arr[], int n) {
    // Phase 1: Detect that a cycle exists
    int slow = arr[0];
    int fast = arr[0];

    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);

    // Phase 2: Find the entry point of the cycle (the duplicate number)
    int slow2 = arr[0];
    while (slow != slow2) {
        slow = arr[slow];
        slow2 = arr[slow2];
    }

    return slow;
}

int main() {
    int n;
    // Read the number of elements
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    // Read the array elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Find and print the duplicate
    int duplicate = findDuplicate(arr, n);
    printf("%d\n", duplicate);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1, n2;

    // Read size of the first array
    if (scanf("%d", &n1) != 1) return;
    int arr1[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    // Read size of the second array
    if (scanf("%d", &n2) != 1) return;
```

```c
    int arr2[n2];
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1; // Flag to handle spacing in output

    // Two-pointer traversal
    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
        } else if (arr1[i] > arr2[j]) {
            j++;
        } else {
            // Found a common element
            if (!first) {
                printf(" ");
            }
            printf("%d", arr1[i]);
            first = 0;

            i++;
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while (t--) {
        solve();
    }

    return 0;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1, n2;

    // Read the size and elements of the first array
    if (scanf("%d", &n1) != 1) return;
    int arr1[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    // Read the size and elements of the second array
    if (scanf("%d", &n2) != 1) return;
```

```
    int arr2[n2];
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1; // Tracks spacing between output elements

    // Linearly traverse both sorted arrays using the two-pointer technique
    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
        } else if (arr1[i] > arr2[j]) {
            j++;
        } else {
            // Match found
            if (!first) {
                printf(" ");
            }
            printf("%d", arr1[i]);
            first = 0;

            i++;
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while (t--) {
        solve();
    }

    return 0;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Pranavkrishna V S 250701540-2025-CSE P2

Started on	Tuesday, 25 August 2026, 10:32 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 10:34 AM
Time taken	2 mins 47 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int hasPairWithDifference(int arr[], int n, int k) {
    int i = 0;
    int j = 1;

    // Search for the pair using two pointers
    while (i < n && j < n) {
        // Ensure indices are distinct (i != j)
        if (i == j) {
            j++;
            continue;
        }

        int diff = arr[j] - arr[i];

        if (diff == k) {
            return 1; // Pair found
        } else if (diff < k) {
            j++; // Difference is too small, increase the larger number
        } else {
            i++; // Difference is too large, increase the smaller number
        }
    }

    return 0; // No such pair exists
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

```

```
    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) return 0;

    // Check for the pair and print the result
    int result = hasPairWithDifference(arr, n, k);
    printf("%d\n", result);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

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Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int hasPairWithDifference(int arr[], int n, int k) {
    int i = 0;
    int j = 1;

    // Search for the pair using the two-pointer approach
    while (i < n && j < n) {
        // Ensure indices are distinct (i != j)
        if (i == j) {
            j++;
            continue;
        }

        int diff = arr[j] - arr[i];

        if (diff == k) {
            return 1; // Pair found
        } else if (diff < k) {
            j++; // Difference is too small, advance the fast pointer to get a larger num
        } else {
            i++; // Difference is too large, advance the slow pointer to decrease the gap
        }
    }

    return 0; // No valid pair exists
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

```

```
    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) return 0;

    // Evaluate pair existence and output the result
    int result = hasPairWithDifference(arr, n, k);
    printf("%d\n", result);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

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